Lum Yu Wei, Edric

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SUMMARY

A third-year student in Singapore University of Technology and Design, currently pursuing a Bachelor's in Engineering Product Development. Interested in designing and solving problems related to automation and robotics, with a strong eagerness to learn more in the field.

WORK EXPERIENCE

CEVA Logistics - Robotics and Automation Intern **Aug 26 – Expected Jan 26**

- Operate and troubleshoot automated kitting machine involving KUKA SCARA robotic arm, conveyor system, PLC, and HMI.
- Design and 3D-print custom conveyor components, including smooth-transition curved guide tracks as well as wheel rollers optimized for 90-degree merges.
- Troubleshoot and fabricate parts to assist in the operation of a label applicator machine, coordinating with the manufacturer for technical resolution.

Griffin Labs - Software Intern **Sep 25 - Jan 26**

- Set up and co-developed an embodied AI robot, the Lionsbot T1, 1 of 3 nominees for the Amsterdam Innovation Award at Interclean 2026, with the team at Griffin Labs, during which the following below was done.
- Configured robust hardware integration, including developing functional Unified Robot Description Format (URDF) models with realistic inertial properties for a 4-wheel swerve drive robot with two arms.
- Wrote ROS2 packages for odometry and control of a 4-wheel swerve drive, ensuring proper interfacing with the motors' driver libraries.
- Co-developed solutions with colleagues for unique problems, including the alignment of a 4-wheel swerve drive robot to fixtures identified via computer vision and AI models using geometry and alignment algorithms.
- Assessed and completed setup of multiple platforms for Nvidia Jetson Orin and Thor, including building drivers for gs_usb and CH341.

Singapore Polytechnic - Temporary Staff **Feb 22 - Aug 22**

- Documented and provided comprehensive write-ups and training of personnel on the ROS2 Navigation stack, covering critical modules such as odometry, basic path planning, and mapping, reverse engineering the Robotis Turtlebot3 Waffle Pi.
- Devised a custom TCP/IP to ROS2 bridge with a graphical user interface (GUI) using QT libraries, facilitating seamless communication between disparate systems.

EDUCATION

Singapore University of Technology and Design (SUTD) **Singapore**

Bachelor's degree in Engineering Product Development **Sep 24 – Expected Apr 28**

- Soldered and wired a full robotic system, involving an Ackermann steering system using linear actuators and motors, ESP32, Raspberry Pi 5 and camera. Trained a YOLO26 model and interfaced ESP32, Raspberry Pi 5 and camera components using ROS2 and UART.

Singapore Polytechnic **Singapore**

High School Diploma in Mechatronics, Robotics, and Automation Engineering **Jan 19 - Feb 22**

- Programmed an autonomous guided vehicle with three other students using ROS2, as well as assisted in the building process.

School of Science and Technology, Singapore **Singapore**

O Levels **Jan 15 - Dec 18**

- Strong emphasis on collaborative work, with numerous projects accomplished in teams, honing teamwork and resilience.
- Competitive Robotics experience, including third place and an innovation award in international competitions.

LEADERSHIP EXPERIENCE

SP Robotics Innovation & Technology Enterprise, Project Coordinator **Jan 20 - Dec 20**

- Completed multiple events as a team, including C++ workshops and robotics projects, as well as managed and assisted in live streaming of projects during the COVID-19 period for open house.

PROJECTS

- **Scoopbot**: Engineered the electrical wiring, safety architecture (E-stop, fuses), and hardware integration for an autonomous waste-collection robot, leveraging ROS2, a remote RTX 4060 laptop computer node for YOLO-based object detection, and a real-time video streaming pipeline for teleoperation, aiming to create a subsystem working in conjunction of current systems to prevent the smearing of fecal matter by current mop/brush systems on automation. Attained best prototype award. https://drive.google.com/file/d/1zo1Uyya581S1UAfItDmF1y3HeIX7xR6O/view?usp=sharing
- **Autonomous Guided Vehicle**: Learned ROS2 from scratch, modifying existing packages to work with a holonomic drive autonomous vehicle to transport objects from point to point. https://drive.google.com/file/d/1_22B_G978wlpDhq0FP9efLj7pabYodQW/view?usp=sharing
- **Computer Numerical Control Plotter**: A passion project with friends, researched and identified libraries to assist and help in the operation of the h-bot system. https://drive.google.com/file/d/1mLr23n5xCTd49cw4pKoblPryRz5_FKzw/view?usp=sharing